CostVision
AI COST INTELLIGENCE

Agentic AI in CostVision

A costing tool that learns from every analysis, remembers your parts, and finds savings on its own — while staying fully auditable.

Remembers · **Recognises** · **Self-corrects** · **Acts autonomously**

Management briefing · July 2026 · All figures verified from live system runs

EXECUTIVE SUMMARY

What we built — in one slide

36×

Error reduction

Estimating error fell from 10.9% to 0.3% after the tool learned from just 3 real quotes (verified live).

£512k/yr

Found autonomously

In our live demo the background agent flagged £512k/yr of pricing issues — with nobody at the keyboard.

99%

Part recognition

A new bracket was matched to 3 past bracket analyses at 98–99% similarity, with reasons shown.

1,005

Automated tests

Every capability is covered by automated tests (86 suites), exercised end-to-end on the running system — and the container build is CI-verified.

The idea, in plain words

Until now, every costing started from zero and the result depended on who did it. **Now the tool keeps an organisational memory:** every analysis is stored, every real supplier quote teaches it, and every new part is compared against everything we have costed before.

The knowledge stays in our database, on our servers — it becomes a company asset that gets more valuable with use, and it does not walk out of the door when an expert leaves.

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THE CONCEPT

What "Agentic AI" means here — four plain words

Remembers

Every costing is saved as a "case": the part, the inputs, the result, and any real quote. Shared across the whole team.

Recognises

Start a new part and it instantly finds the most similar past parts — like an experienced engineer saying "we've done this before".

Self-corrects

Log the real supplier price and the tool measures its own error, then corrects future estimates in that category. Accuracy is measured, not claimed.

Acts

A background agent re-checks all stored parts on a schedule and raises findings by itself: "you are overpaying here — worth £400k/yr."

Deliberate design choice: the learning is statistics over our own data — every suggestion shows its source parts and its arithmetic. That makes it auditable and defensible in front of a supplier, which is where costing tools win or lose.

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HOW IT WORKS

How it works — the learning loop

1 · **Analyse** Engineer costs a part as usual
2 · **Remember** Saved automatically to the knowledge base
3 · **Recognise** Similar past parts found instantly
4 · **Suggest** Benchmarks, materials, real quotes shown
5 · **Learn** Real quote logged → model self-corrects
6 · **Act** Background agent flags drift & savings

…and every loop makes the next estimate better

No extra work for the engineer. Steps 2, 3, 4 and 6 happen automatically. The only new habit is one click — "Log Actual £" — when a real supplier quote arrives. That single click is the fuel for everything on this slide.

CAPABILITY 1 OF 5

The memory — an organisational knowledge base

What is stored for every analysis

- **Part "fingerprint" —** process, material, weight, size, region, volume
- **The full cost result —** total + breakdown by cost driver
- **Real quotes —** actual supplier / PO prices, when logged
- **Expert corrections —** where a person adjusted the AI's values
- **CAD shape data —** dimensions & features, when a CAD file was used

Why it matters to us

✓ **Shared, not personal**
One engineer's analysis instantly helps everyone — juniors inherit senior judgement.

✓ **Stays on our servers**
Our database, our infrastructure. The knowledge is a company asset, not a vendor's.

✓ **No duplicates**
Re-costing a part updates its case — the memory stays clean.

✓ **Compounds with use**
Useful from ~20–30 analyses; every costing from today is an investment.

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Recognition — "we have costed this before"

Live example: an engineer costs a new 0.85 kg aluminium bracket. The tool answers instantly:

🧠 **AI memory — similar past parts** (knowledge base: 3 analyses · 2 with actuals)

EV Battery Bracket	99% match — material family, weight, region	**£41.20**	**actual £46.50**
Sensor Mount Bracket	98% match — material family, weight, region	**£38.50**	
Inverter Bracket	98% match — material family, weight, region	**£44.80**	**actual £50.20**

- Median cost of 3 similar parts: £41.20
- 3 of 3 used the same material (aluminium 6061)
- Real quotes logged for 2 of them — median actual £48.35

Every match is explained (what matched, and how strongly) and every suggestion names its source parts — **no black box.**

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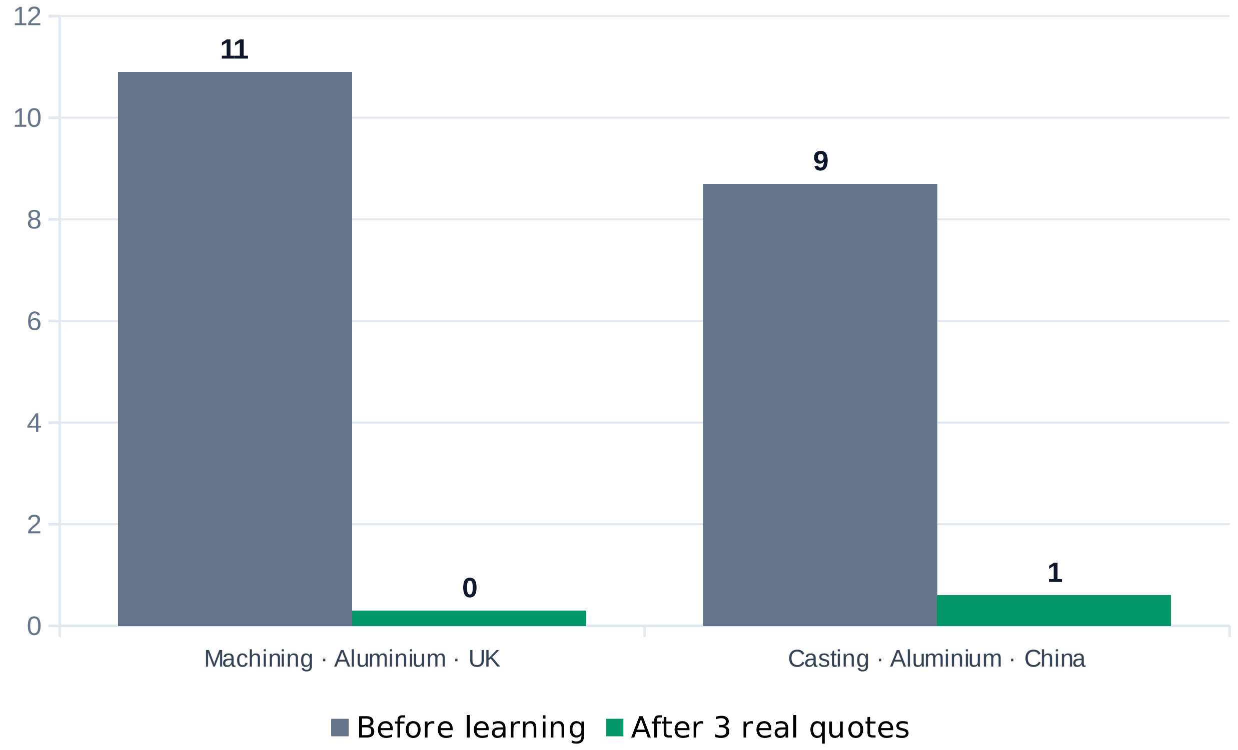
CAPABILITY 3 OF 5

Self-correction — it learns from real quotes



Estimating error (%) — measured against real supplier prices, live system run

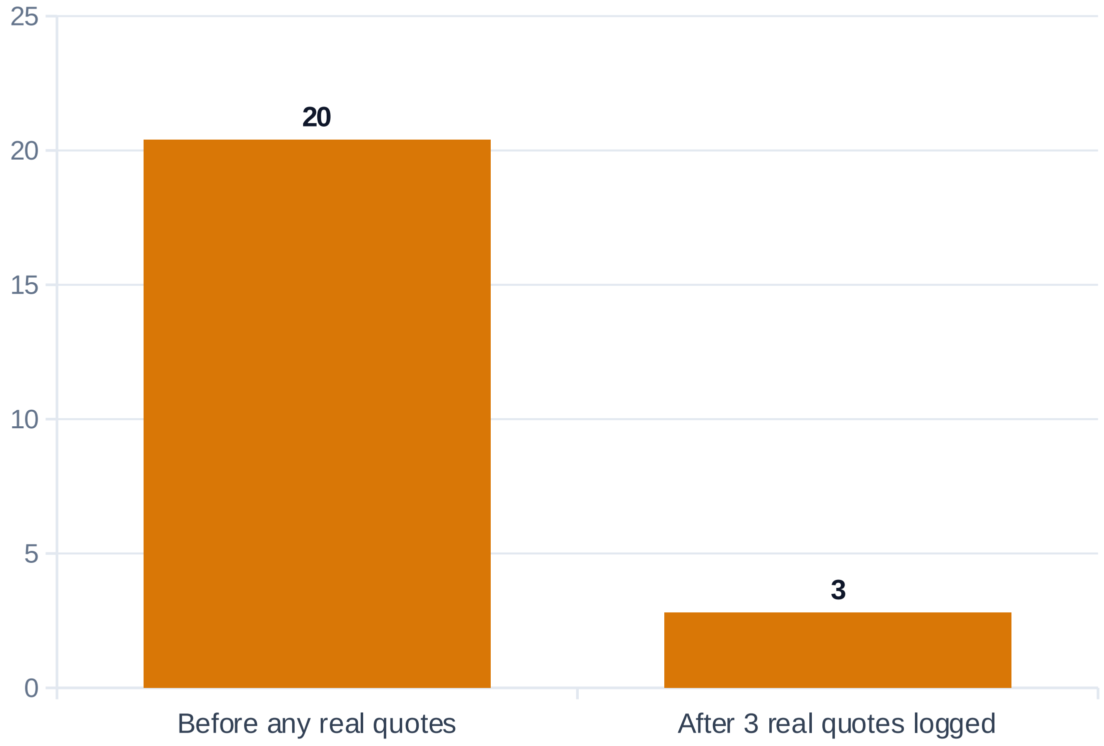
How it works, simply

1. A real supplier quote arrives → one click logs it.
2. The tool compares its estimate with reality and measures its own error.
3. From 3 quotes in a category, it corrects future estimates automatically — per process, material family and region.
4. It reports its accuracy openly (error % before and after).

Why per-category matters: our machining ran 12% low while our China castings ran 8% high — averaged together they looked "fine". The tool catches what averages hide.

CAPABILITY 4 OF 5

Honest ranges — from "±20%" to "±3%", earned with data



Width of the cost confidence band on the same part (± % around the estimate)

Every estimate now comes as a range

Optimistic (P10) · Most likely (P50) · Conservative (P90)

A single number implies a precision that early estimates don't have. The range tells buyers how much to trust the number — and what target to set in negotiation.

The band is earned, not guessed: it starts wide when the tool has no evidence, and tightens automatically as real quotes prove the accuracy. On our test part it went from ±20% to ±3% after three quotes.

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CAPABILITY 5 OF 5

The autonomous agent — finds money unattended

A background monitor re-checks every stored part on a schedule. In the live demonstration it opened these findings **entirely unattended:**

💰 **RENEGOTIATION** — **≈ £400,000 / yr**

Supplier price £48.00 is 20% above should-cost £40.00 at 50,000 pcs — clear recovery opportunity.

🚨 **UNDERWATER PRICE** — **≈ £112,000 / yr exposure**

Supplier price 25% BELOW should-cost — verify scope and quality; pricing may be unsustainable.

⏳ **STALE ESTIMATE** — **Confidence decay**

A 120-day-old estimate was never validated with a real quote — the agent asks for a refresh.

Total surfaced in the demo: **≈ £512,000 / yr** — every finding shows its arithmetic, and one click dismisses it once handled.

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THE WIDER AI PLATFORM

Supporting AI capabilities already in the tool

📚 Grounded AI assistant

Answers cite our actual rate library and past costings — the AI quotes our data, it doesn't improvise. Every figure is traceable.

📄 RFQ analyst

Drop in an RFQ package: it costs every line, flags risky prices and single-source parts, and drafts a prioritised negotiation brief.

📐 CAD feature costing

Reads the 3D model and prices individual design features — holes, threads, surfaces — so designers see exactly what drives cost.

🌱 Carbon co-costing

Every part gets a CO_2e figure alongside the £ — increasingly demanded in automotive & aerospace RFQs.

Together with 18 commodity cost engines, CAD-to-cost, and PCB photo-to-BOM — the agentic layer sits on top of all of it.

INPUTS REQUIRED

What it needs from us — honestly, very little

You provide

1. **Keep costing parts as normal**
   every analysis feeds the memory automatically
2. **One click when a real quote arrives**
   "Log Actual £" — this is the learning fuel
3. **Optional: CAD file or BOM**
   sharper matching and better auto-fill
4. **Optional: historic quotes (CSV)**
   seeds the memory so it starts smart, not empty

The tool does

- ✓ Remembers every analysis (no duplicates, org-wide)
- ✓ Finds & explains similar past parts instantly
- ✓ Suggests benchmarks, materials and real prices
- ✓ Measures its own error and self-corrects per category
- ✓ Tightens confidence ranges as evidence grows
- ✓ Monitors all parts in the background & raises findings

EVIDENCE

Results & accuracy — measured, not promised

Estimating error after learning (machining segment)	**10.9% → 0.3%**
Estimating error after learning (casting · China)	**8.7% → 0.6%**
Confidence band on the same part	**±20.4% → ±2.8%**
Similar-part recognition on live example	**98–99% match, reasons shown**
Autonomous findings in unattended demo	**£512,000 / yr surfaced**
Automated tests protecting all of this	**1,005 passing (86 suites)**

How we verified: every number above comes from running the real system end-to-end — live server, real database, real API calls — not from slides or simulations.

BENEFITS

What this means for the business

⚡ Faster costing

New parts start from proven history instead of a blank sheet — matches, materials and benchmarks appear instantly.

🎯 Accuracy that compounds

Every real quote makes the next estimate better. Accuracy is measured and reported — defensible in any negotiation.

🏦 Knowledge stays

Senior engineers' judgement is captured as data. It doesn't leave when people do — and juniors inherit it from day one.

💰 Money found proactively

The agent watches all parts continuously and flags overpayment and pricing risk by itself, quantified in £/yr.

And it is ours: the knowledge base runs on our infrastructure and grows into a proprietary asset competitors cannot buy.

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Confidence you can defend — not just assert

Every should-cost now carries two ranges — the physics estimate, and an **empirical band proven against your own logged quotes.**

Physics prior (Monte-Carlo)

How well the inputs are known — before you have any real quotes.

Example: £86.34 ± 3.4%

Always available, every part, every commodity.

Empirical band (conformal)

Proven against the quotes YOU logged — with a coverage guarantee.

Example: 90% of your quotes land within ± 6.5% → £81.54 – £92.88

Tightens automatically as more quotes are logged.

Why it helps: a buyer can’t defend "trust me, ±5%". They CAN defend "90% of our real quotes for this family landed within ±6.5%." The band edge is an observed error — evidence, not an assertion. It is the honest uncertainty no competitor states this way.

NEW IN 2026 · ADVANCED INTELLIGENCE

The agent learns what actually earns money

The autonomous agent no longer just finds gaps — it learns which findings **actually convert into savings, and re-ranks by the money it can really recover.**

Finding	Raw gap (impact)	Learned conversion	Expected realizable
Cast Housing	£200k/yr	20% — rarely closes	**£40k**
Machined Knuckle	£100k/yr	80% — usually closes	**£80k**

The result: the machining finding — HALF the raw gap — now ranks ABOVE the casting one, because the agent learned casting renegotiations don’t close. It stops shouting about theoretical money and surfaces the money you can get.

Why it helps: sourcing time is scarce. The agent spends it where the return is real — and tracks the £ actually saved, so you can prove the tool paid for itself. One click ("Actioned £") teaches it after every negotiation.

NEW IN 2026 · ADVANCED INTELLIGENCE

The negotiation coach — it hands you the argument

The tool now knows ***why*** a part costs what it does — and turns that into the sentence that wins the negotiation.

Live example — real output from the tool

“Material is £8.39 of this part, driven by Aluminium. Every 1% move in the Aluminium index shifts the piece price by £0.11. A quote of £95 is only justified if Aluminium were ~14% above today’s index — ask the supplier to show that, or hold at £86.34.”

What it does

Links the material to its commodity index, then works out — through the same maths the engine uses — exactly how much a supplier’s price implies the metal has moved.

Why it helps

The buyer walks in with a defensible, arithmetic counter-argument instead of "that feels high". It converts should-cost into negotiating power — the number and the words to win with it.

NEW IN 2026 · ADVANCED INTELLIGENCE

Cost weather on demand — the what-if engine

Ask "what if a commodity moves?" and get the answer instantly — for one part, or the whole portfolio.

One part — the live slider

Drag a commodity −20% … +20% and the piece price recomputes as you move.

Example: Aluminium +15% → £86.34 → £87.92

Instant sensitivity in a single gesture.

Whole portfolio — the scenario

Apply a move across every part at once and see who changes status.

Example: "If Steel +10% → 7 parts cross underwater, £X/yr at risk."

Pre-empt losses before the market moves them.

Honest by design: this is a conditional — "IF the index moves" — never a price forecast. So it stays defensible even on an indicative commodity feed, and upgrades to a true forecast the day a live market feed is connected.

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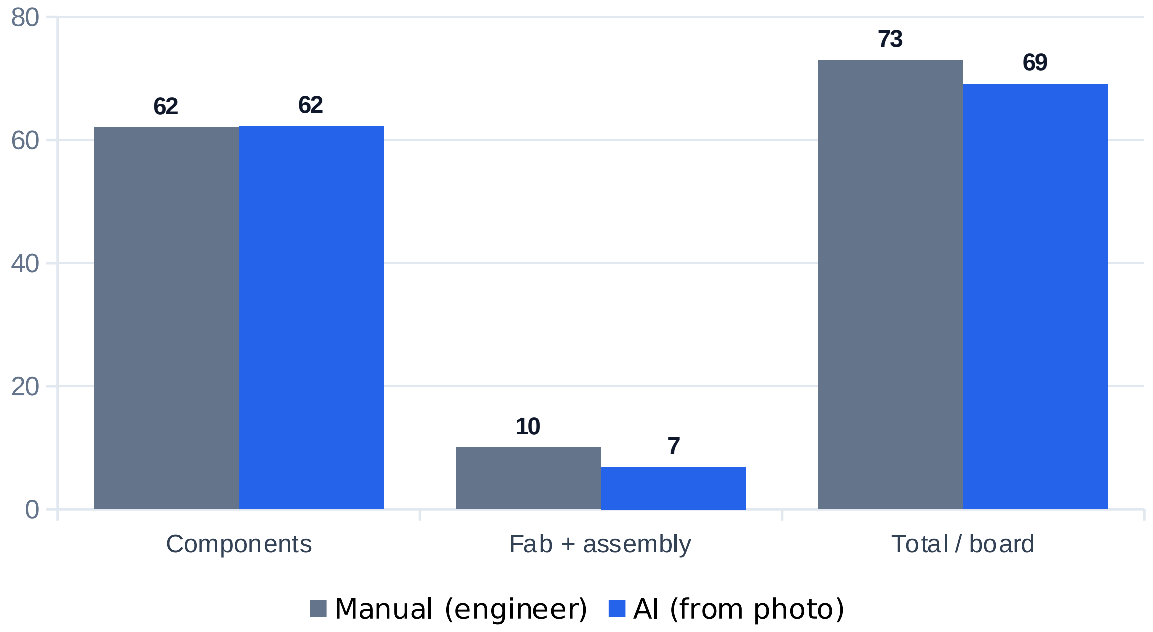
PCB photo → BOM → should-cost — now supplier-grade

Photograph a circuit board, get a costed bill of materials in ~60 seconds. This year we hardened it from a clever demo into a **number you can put in front of a supplier.**



Real automotive ECU, China @ 10k/yr — AI landed within ~5% of a half-day manual

What we fixed this year

- ✓ **Empty-BOM bug — killed —** Complex boards used to return nothing; now we read every block the model emits.
- ✓ **Catalogue grounding —** Confirmed parts snap to real market prices — offline, no external API.
- ✓ **Class-median cap + magnitude guard —** A misread part can no longer dominate: one MCU £84 → £18, capped & flagged.
- ✓ **Deterministic fabrication —** Fab is derived from stable board features — the headline no longer swings run-to-run.
- ✓ **Confirmed vs needs-verification —** The headline splits the £ you can trust from the £ to firm up.

Result: the 2–3× over-costing is gone. That live ECU came back ASIL-B, 23 BOM lines, £69.11/board — with £37.65 confirmed and £24.64 honestly flagged to verify. Same photo, same answer, every run.

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The tool now checks its own homework

Every lesson we learned fixing real parts became a **deterministic check that runs on every single estimate** — catching a wrong machine, an implausible wall, or a mis-set weight before it ever reaches a supplier.

✓ **Machine sized to the part**	Flags an under-sized press or mould and offers the right one in one click
✓ **Geometry sanity**	A wall thicker than the part, or a thin hollow shell mis-read as solid — caught
✓ **Weight vs. geometry**	Costed weight must match the measured volume × the chosen material's density
✓ **Tooling amortised right**	Tooling must amortise over the stated annual volume — every commodity, no exceptions

Live proof: we deliberately put a forging on a 2-tonne hand-hammer; the tool flagged it HIGH, named the right 1,600-tonne press, and one click restored it. It re-derives from physics — so it also catches a colleague's override or a future code regression, not just yesterday's bug.

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NEW IN 2026 · ADVANCED INTELLIGENCE

Proven on five real automotive CAD parts

Five real STEP files, run end-to-end. On each, the tool's first answer had a specific, explainable error — every one is now fixed and guarded by a test.

Part	First answer	Corrected	What it caught
Fuel tank	£216.97 sand casting	**£24.62 blow-moulded**	thin hollow shell ≠ a metal casting
Front bumper	£29.18 Al casting	**£7.79 injection-moulded**	plastic mis-read as metal; 27 mm phantom wall
Servo horn	£333 machining card	**£6.40**	266 min of cutting on a 3 g part
Stub axle	wrong — lightest metal	**forged steel, 8.1 kg**	picked the lightest material, not the real one
Seat cross-member	crash — missing field	**£1.11 steel stamping**	a missing AI field must never break the calc

The point: geometry is ground truth; the AI only classifies material and process — and when it slips, the guardrails now catch it. Every fix shipped with a test, so the same mistake can't come back.

THE DIFFERENTIATOR

Why ours is different — glass-box autonomy

The principle: every learned or derived number stays auditable — a value a cost engineer can read and defend. No black-box weight ever touches the price. The AI narrates and explains; it never decides the number in secret.

Continuous learning	Calibrates on YOUR quotes; conformal band with a guarantee
Autonomous action	Agent opens findings unattended AND learns which convert
Causal reasoning	Knows why a part costs what it does; coaches the negotiation
Explainable — always	Every number defensible line-by-line; the competitor’s edge is the opposite
Runs in your walls	On-premise; the knowledge is your IP and never leaves

NEXT STEPS

Where we are, and the ask

Status today

- ✓ All capabilities built, tested (1,005 tests) and live — incl. 8 new for 2026
- ✓ Verified end-to-end; ships as a container (CAD engine included), CI build-tested
- ✓ Zero extra licence cost — built into our tool
- ✓ Runs on-premise; no data leaves the company

Honest note: the intelligence starts empty and grows with use. The mechanism is proven; the value builds as we feed it.

The ask — three small decisions

1. **Adopt the habit**
   Make "Log Actual £" part of the quote-handling routine. One click per quote.

2. **Seed the memory**
   Approve a one-off import of historical quotes so the tool starts smart (~1–2 days of effort).

3. **Review the findings**
   Put the agent's monthly findings on the sourcing team's agenda — it is already finding money.